QR Ordering Platform — Build Documentation

Nov 13, 2026 · @Kushwant KR

Start here

This set specifies an original QR-based restaurant ordering platform, modelled on the feature set described in the Qrder case-study page. Six tabs, read in this order:

#	Tab	What it settles	Feed to the build agent when
1	PRD (this tab)	What to build, for whom, and what is out of scope	Always, first
2	Architecture	Stack, data model, API, realtime, security	Scaffolding and every backend task
3	Design System	Tokens, components, screen patterns	Every UI task
4	AI Agents Guide	Rules, repo layout, task order, definition of done	Loaded as the agent's standing instructions
5	Testing Guide	Test layers, critical cases, release gates	Every task, before marking done
6	Application Flow	Screen-by-screen journeys and the order state machine	Any feature touching routing or order state

Three ground rules for the build

1. Build the capability, not the brand. Use your own product name, logo, copy, colours and screenshots. Do not reuse the name Qrder, Pegorion's wording, or their imagery.
2. Every requirement below is tagged **Observed** (stated on the source page) or **Inferred** (my design decision to fill a gap). Inferred items are yours to change.
3. The source page's results (45% faster service, 30% lower cost, 22% higher basket, 4.9/5, ₹5–7 lakh saved) are their unverified marketing claims. Treat them as targets to measure, never as claims for your own product.

Source analysis

The source is a marketing case study, not the application, so it reveals the feature surface but nothing about the implementation. I found no public technical write-up of the product in a web search.

What the page states (Observed)

Area	Evidence on the page
Positioning	Hospitality tech, QR-based ordering, enterprise SaaS, software-only deployment
Guest flow	Scan a table QR, view the menu, place an order, pay from the phone, no app install
Menu	Images, dietary filters, upsells ("smart upsells" raising basket size)
Orders	Live order tracking, orders grouped by table, automated routing to the kitchen
Owner tools	One dashboard for menus, orders, tables, payments; billing and inventory mentioned
Kitchen	Dedicated kitchen screen with clear, sequential order instructions in real time
Integrations	"Seamless POS integration", integrated payment methods
Insights	"No insights" is listed as a problem solved, implying analytics
Market	Prices in ₹, so India: UPI, GST
Four screens shown	Restaurant Dashboard, Student Ordering Interface, Canteen Management Panel, Kitchen Screen

What that implies

- The "Student" and "Canteen" screens show the product runs in two modes: **dine-in** (order tied to a table, served at the table) and **canteen/counter** (order tied to a token, collected at a counter). The data model must support both.
- "Enterprise SaaS" implies multi-tenancy: many restaurants, each with outlets, staff roles and isolated data.

What the page does not tell us (all Inferred in these docs)

- Tech stack, database, hosting, realtime mechanism
- Visual design: colours, type, spacing (only text was captured, no screenshots)
- Payment provider, POS vendors supported, pricing model
- Auth model for guests and staff, offline behaviour, printing

If you can share screenshots of the live preview, I can tighten the Design System and Application Flow tabs against what is actually shown.

Product Requirements Document

1. Product summary

A multi-tenant web platform that lets guests scan a QR code, order and pay from their phone with no app install, while staff run the kitchen, floor and menu from live dashboards. It serves two venue modes: dine-in restaurants (table QR) and canteens/counters (token pickup).

2. Problems to solve

Problem (Observed)	Product response
Slow response: guests wait to order and to pay	Self-serve ordering and payment from the table
Inaccuracies: manual order entry causes waste	Guest-entered orders routed straight to the kitchen
Labour gaps: high turnover hurts service	Fewer order-taking and billing touchpoints
No insights: no data for staffing or menu decisions	Sales, item and peak-hour analytics

3. Users and roles

Role	Who	Main jobs	Auth
Guest	Diner or student	Browse, order, pay, track	None; session bound to a QR token. Optional phone number for receipts

Role	Who	Main jobs	Auth
Owner	Restaurant owner	Set up outlet, menu, tables, staff; see reports; billing	Email + password or OTP
Manager	Outlet or canteen manager	Run live orders, edit menu availability, handle refunds	Same as Owner, scoped to outlet
Kitchen	Cooks, expediter	See the queue, advance order status	Shared device login or PIN
Waiter/Counter	Floor or counter staff	Serve, mark delivered, take cash, assist guests	PIN
Platform admin	You (SaaS operator)	Onboard tenants, plans, support	SSO, separate admin app

4. Scope

MVP (must ship)

- Tenant, outlet and staff setup with role-based access
- Menu management: categories, items, images, price, veg/non-veg and dietary tags, variants, add-ons, availability toggle
- Table management and QR generation (print-ready PDF per table)
- Guest web app: menu, search, filters, cart, order notes, place order, live status
- Kitchen display screen (KDS) with realtime queue and status changes
- Live orders board grouped by table, with running bill per table
- Payments: UPI and cards through one gateway, plus "pay at counter" cash
- GST-compliant bill and digital receipt
- Canteen mode: token number, ready-for-pickup notification
- Basic analytics: sales by day, top items, peak hours, average order value

Phase 2

- Upsell and combo suggestions in cart (Observed as "smart upsells")
- Inventory: stock counts, auto mark-unavailable, low-stock alerts (Observed as a claim)
- Split bill, tips, coupons, loyalty
- KOT and bill printing (thermal printers)
- Guest ratings and feedback
- Multi-language menus

Phase 3

- POS integration (Observed; vendors unknown), aggregator integrations
- Multi-outlet rollups, staff scheduling insights, demand forecasting
- Subscription billing for tenants, white-label domains

Out of scope

- Delivery logistics, table reservations, native mobile apps, payroll

5. Functional requirements

Priority: P0 = MVP blocker, P1 = MVP should, P2 = later. Source: O = Observed, I = Inferred.

5.1 Guest ordering

ID	Requirement	Pri	Src
G-01	Scanning a table QR opens the outlet menu in the browser with table pre-selected; no install, no login	P0	O
G-02	Menu shows categories, item image, name, description, price, veg/non-veg mark, dietary tags	P0	O
G-03	Guest can search and filter by dietary tag and category	P0	O
G-04	Item sheet supports variants (size) and add-ons with min/max selection rules	P0	I
G-05	Cart persists across refresh for the session; shows taxes and total before placing	P0	I
G-06	Guest can add a note per item and per order	P1	I
G-07	Placing an order returns an order number and opens the live tracking screen	P0	O
G-08	Status updates appear within 2 seconds of a staff change, without refresh	P0	O
G-09	Guest can place further orders on the same table; all join one running bill	P0	I
G-10	Guest can pay online (UPI, card) or choose pay at counter, per outlet setting	P0	O
G-11	Guest receives a receipt on screen, downloadable, optionally by SMS/WhatsApp	P1	I

ID	Requirement	Pri	Src
G-12	Unavailable items show as sold out and cannot be added	P0	I
G-13	Cart suggests add-ons or pairings	P2	O
G-14	Canteen mode: guest gets a token number and a "ready" alert	P0	O
G-15	Guest can call a waiter or request the bill	P1	I

5.2 Kitchen screen

ID	Requirement	Pri	Src
K-01	New orders appear in realtime, oldest first, with table or token, items, variants, notes	P0	O
K-02	Staff advance a ticket: New, Preparing, Ready; single tap, large targets	P0	O
K-03	Audible and visual alert on new order	P0	I
K-04	Ticket shows elapsed time and turns warning/overdue at configurable thresholds	P1	I
K-05	Items can be marked done individually	P1	I
K-06	Screen reconnects and resyncs after network loss without losing tickets	P0	I
K-07	Optional station filter (grill, drinks)	P2	I

5.3 Restaurant dashboard

ID	Requirement	Pri	Src
D-01	Live orders board grouped by table, with status and running total	P0	O
D-02	Staff can accept, reject (with reason), cancel, and mark served	P0	I
D-03	Table map/list shows free, occupied, bill requested	P0	O
D-04	Menu CRUD with image upload, drag-to-reorder, bulk availability toggle	P0	O

ID	Requirement	Pri	Src
D-05	Table CRUD and QR generation, with regenerate to invalidate old codes	P0	O
D-06	Close table: settle bill by online, cash or mixed; generate GST invoice	P0	O
D-07	Payments list with status, refunds through gateway	P1	O
D-08	Staff management: invite, role, PIN, deactivate	P0	I
D-09	Outlet settings: hours, taxes, service charge, mode (dine-in/canteen), payment options	P0	I
D-10	Analytics: sales, orders, AOV, top/bottom items, peak hours, prep time	P1	O
D-11	Inventory tracking linked to items	P2	O

5.4 Canteen panel

ID	Requirement	Pri	Src
C-01	Incoming order queue by token, with paid/unpaid flag	P0	O
C-02	Quick menu availability edits during service	P0	O
C-03	Mark ready and handed over; ready tokens show on a public display view	P1	I
C-04	Prepaid-only setting to prevent unpaid pickups	P1	I

5.5 Platform admin

ID	Requirement	Pri	Src
A-01	Create, suspend and configure tenants and plans	P1	I
A-02	Impersonate a tenant for support, with audit log	P2	I

6. Non-functional requirements

Area	Requirement
Performance	Guest menu interactive in under 2.5 s on a mid-range Android phone over 4G; guest JS bundle under 200 KB gzipped
Realtime	Order events delivered in under 2 s at p95
Availability	99.9% monthly for ordering paths
Scale (MVP target)	200 outlets, 50 concurrent orders per outlet at peak
Security	Tenant isolation enforced in the database; no price or total trusted from the client; payment webhooks signature-verified
Privacy	Collect guest phone number only when needed; comply with India's DPDP Act
Compliance	GST invoice fields; no card data stored (gateway-hosted checkout)
Accessibility	WCAG 2.1 AA on guest app
Devices	Guest: mobile browsers from the last 3 years. KDS: tablet landscape and TV. Dashboard: desktop and tablet
Resilience	Guest cart survives refresh; KDS survives short network drops
Localisation	English at MVP; currency ₹; structure ready for Hindi and Telugu

7. Success metrics

Measure these per outlet from day one; the source page's numbers are the benchmark to beat, not a promise.

Metric	Definition	Target
Scan-to-order conversion	Sessions placing an order / QR sessions	60%
Time to first order	QR scan to order placed, median	Under 3 min

Metric	Definition	Target
Order-to-ready time	Placed to Ready, median	Baseline, then down 20%
Order error rate	Orders cancelled or remade for wrong items	Under 1%
Online payment share	Orders paid online / all orders	70%
Average order value	Revenue / orders	Up 10% after upsells
Payment success rate	Successful / attempted	95%

8. Milestones

Milestone	Contents	Exit test
M0 Foundation	Repo, auth, tenant model, RLS, CI	Two tenants cannot read each other's data
M1 Menu and tables	Menu CRUD, tables, QR PDF	Owner builds a 30-item menu and prints QRs
M2 Guest ordering	Menu, cart, place order, tracking	Order placed from a phone appears in DB
M3 Kitchen and live board	KDS, orders board, realtime	Status change reaches the guest in under 2 s
M4 Payments and billing	Gateway, webhooks, cash, GST bill	Paid order reconciles; refund works in test mode
M5 Canteen mode and analytics	Tokens, pickup display, reports	Full canteen run-through passes
M6 Pilot hardening	Load test, a11y, monitoring	Release gates in the Testing Guide pass